DATABASE MANAGEMENT SYSTEM

(ENCS205)

Lab Report

Bachelor of Technology in Computer Science and Engineering

Submitted by-

DIYA GOEL (2401010179)

Course Teacher-

Ms. MANSI KAJAL

School of Engineering & Technology

K. R. MANGALAM UNIVERSITY

Sohna, Haryana 122103, India

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Experiment No: 1

Aim: To design an Entity Relationship (ER) diagram for the College database and convert the ER diagram into relational tables.

Description:

The system consists of entities such as Student, Class, Semester, Subject, and Marks. A student attends a class, enrolls in subjects, and scores marks in each subject. Each class belongs to a specific semester, and a semester can have multiple classes. The marks entity stores internal and final assessment details of students.This ER diagram helps in clearly defining relationships between entities and avoids data redundancy. It provides a strong foundation for designing a relational database for managing student academic records.

Entities and Attributes:

1. **STUDENT**

   Represents a student studying in the institution.

   Attributes:

   1) USN (Primary Key)
   2) SName Address
   3) Phone

2. **CLASS**

   Represents a class allotted to students.

   Attributes:

   - USN
   - SSID (Section / Class ID)

2. **SEMESTER**

   Represents the academic semester.

   Attributes:

- SSN
- Sem
- Sec

3. **SUBJECT**

Represents subjects taught in the course.

Attributes:

- Subcode (Primary Key)
- Title
- Credits
- Sem

4. **MARKS**

Represents the marks scored by students.

Attributes:

- USN
- SSD
- Test1
- Test2
- Final
- SCod

Relationships:

STUDENT – CLASS

Relationship: Attends

- A student attends a class.
- A class can have many students.

1 **CLASS – SEMESTER**

Relationship: Belongs To

- A class belongs to one semester
- A semester can have multiple classes.

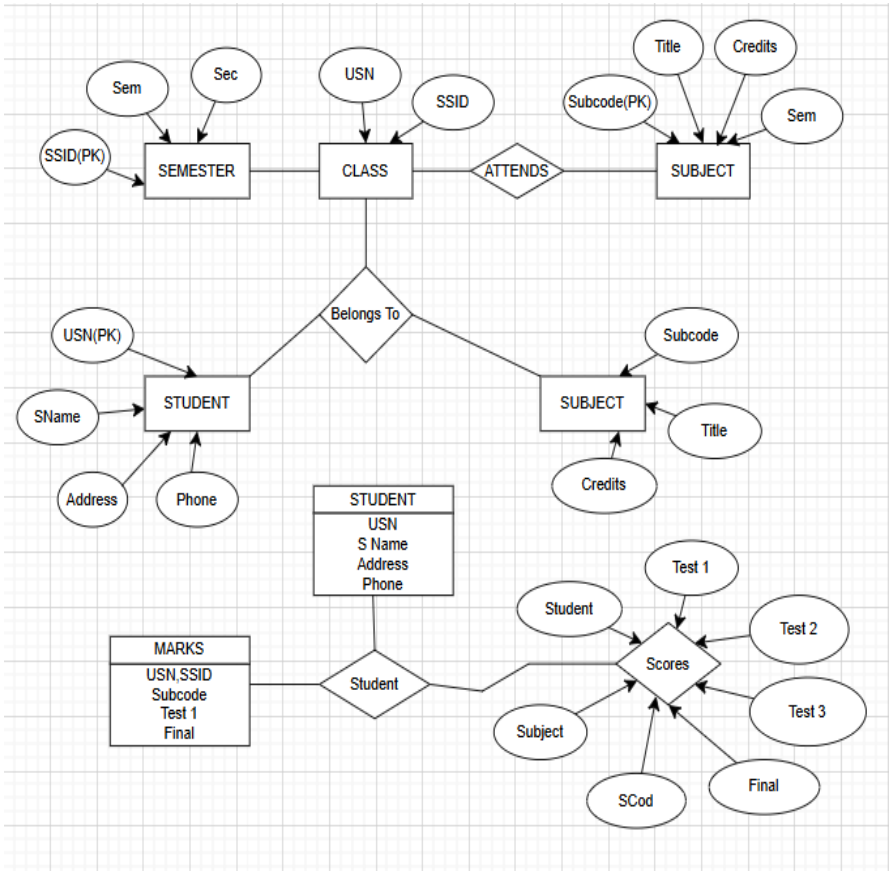
1. **STUDENT – SUBJECT**

   Relationship: Enrolls

   - A student can enroll in many subjects
   - . A subject can have many students.

2. **STUDENT – MARKS**

   Relationship: Scored

   - A student scores marks in different subjects
   - Marks depend on the student and subject.

ER Diagram: